

Sultanate of Oman
Ministry of Health
Royal Hospital



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Title: Policy for Antimicrobial Stewardship Program

Statement

This policy establishes key directives for antimicrobial prescribing and management at the Royal Hospital, with particular reference to antimicrobial prescribing processes, management of antimicrobial formulary and restrictions, use of evidence-based therapy and access to expert clinical advice.

Introduction

Antimicrobial stewardship program (ASP) can be defined as an ongoing and systematic effort to optimize the use of antimicrobial medicines within the hospital. The key objective of this program includes reducing inappropriate antimicrobial use, improving patient care outcomes and mitigating adverse consequences of antimicrobial use (such as antimicrobial resistance, preventable patient harm and unnecessary costs associated with pharmaceuticals expenses and drug-resistant infections).

Scope

This policy applies to Royal Hospital.

Content

The ASP Committee

The Royal Hospital is a multidisciplinary group responsible for the development and ongoing evaluation of ASP. Core membership on this committee consists of representation of disciplines from adult and pediatric infectious diseases, infection control, microbiology, nursing and pharmacy. The ASP team is mainly responsible for providing advice, education, and feedback on antimicrobial prescribing, managing approval requests for restricted antimicrobials and monitoring usage as directed by the ASP plan.

Key Prescribing Principles

All prescribers at Royal Hospital are expected to prescribe antimicrobial therapy according to the following key principals:

1. Therapeutic decisions regarding the prescription of antimicrobials will be based on best available evidence. Empirical antimicrobial therapy (the infective pathogen is not known) or prophylactic therapy (given to prevent acquisition or development of infection) is prescribed according to locally-endorsed guidelines or National Antimicrobial Guidelines (Ministry of Health Booklet, May 2016). When the infective pathogen is known, antimicrobials are to be prescribed according to microbiology results and antimicrobial susceptibilities, where available,
2. Prescribed antimicrobials will be of the narrowest spectrum possible for achieving the intended effect,
3. Dosage, route and frequency of prescribed antimicrobials appropriate for the individual patient, as well as the site and type of infection,
4. The duration of antimicrobial therapy will be defined and/or regularly reviewed (based on evidence-based guidelines and clinical improvement),
5. Monotherapy is used in most indications, where clinically appropriate.

Use of Evidence-Based Prescribing Guidelines and Hospital Guidelines

Locally endorsed hospital guidelines are guidelines that have been reviewed and approved by the ASP Committee, in liaison with the Pharmacy and Therapeutics Committee. If a clinical unit wishes to use treatment guidelines that have not been endorsed by the ASP Committee, these should be submitted to ASP committee for review by the head of the department/unit.

National Antimicrobial Guidelines

The current version of national antimicrobial guidelines should be used in clinical scenarios for which a locally endorsed hospital antimicrobial prescribing guideline does not exist. These guidelines are produced through the directorate general for disease surveillance and control, in the ministry of health. It has been developed by a national multidisciplinary committee of experts. Electronic copy should be available in the Hospital Document system.

Prescribing outside hospital guidelines or national antimicrobial guidelines

If patient is prescribed antimicrobial therapy that is not in concordance with either locally endorsed hospital guidelines or national antimicrobial guidelines, the prescriber is expected to document their clinical reasoning in the patient's notes.

Where an indication or clinical scenario is not covered by either set of guidelines, a prescriber may choose antimicrobial therapy based on an appropriate, peer-reviewed source. Care should be taken to ensure the recommendations made are relevant to the local antimicrobial resistance patterns and availability of the selected antimicrobial agents. A referral should be made for consultation to ASP team or an infectious diseases/microbiology/clinical pharmacy.

Access to Prescribing Resources and Support

Access to Evidence-Based Prescribing Guidelines

All clinical areas have electronic access to antimicrobial prescribing guidelines and national antimicrobial guidelines via the Royal Portal.

Access to Antimicrobial Advice

Prescribers have access to expert advice on antimicrobial prescribing and the management of infectious diseases 24 hours a day, 7 days a week.

The ASP team is the first point of call for general advice regarding antimicrobial prescribing and approval of restricted antimicrobial agents. In more complex cases, the ASP team may recommend a consultation by an infectious disease's physician.

After working hours, prescribers may contact the on-call medical officer for infectious diseases or microbiology.

Local Restrictions on Antimicrobial Prescribing

Formulary and Non-Formulary Antimicrobials

A list of antimicrobial formulary products is available in the Royal Hospital health information system (Al Shifa).

Non-formulary antimicrobials are those not routinely stocked in the ward. Requests for non-formulary antimicrobials will usually require consultation by the infectious diseases team prior to approval, or as a condition of approval by applying using Form A for individual patient and Form B for change of practice which are accessible in the prescribing modules to team consultants. These applications, if not are urgent, are forwarded to the pharmacy and therapeutics committee for approval.

Restricted Antimicrobials

The implementation and maintenance of antimicrobial restrictions is a core ASP strategy at Royal Hospital. The ASP Committee determines the restriction status and access rules for each antimicrobial agent (or specific formulations) based on the following considerations:

- Best practice prescribing
- Patient safety risks associated with particular antimicrobials (including adverse effects, drug interactions, the need for therapeutic drug monitoring, risk of *Clostridium difficile* infection and risk of selecting for more resistant organisms)
- Complexity of relevant disease states and/or public health risks
- Potential for contributing to antimicrobial resistance in the healthcare environment.
- Cost-effectiveness (as compared to alternative agents)
- Resources available to manage antimicrobial restrictions

Prescriber Advocacy and Feedback on Restriction Policy

Feedback from prescribers regarding antimicrobial restriction policy is both expected and encouraged. Royal Hospital prescribers may choose to engage with their representative members of the ASP committee, or may alternatively direct their suggestions or concerns to the ASP Committee Chair or ASP team.

Other Antimicrobial Stewardship Initiatives

ASP Rounds

The ASP team will carry out ASP rounds daily. The purpose of these rounds will be to review antimicrobial prescribing in targeted clinical areas (e.g. critical care, respiratory, hematology/oncology and surgical wards/others) to make recommendations to treating teams, if adjustment of antimicrobial therapy is warranted

Monitoring of Antimicrobial Usage and Resistance

The ASP program incorporates a range of data collection methodologies to monitor both the quantity and quality of antimicrobial usage and examine processes associated with antimicrobial prescribing and supply. These methods include (but are not limited to):

- Analysis of Restricted Antimicrobial Use
- Antibiotic use measure as either days of therapy (DOT) or defined daily dose(DDD)
- Analysis of third generation cephalosporin
- Review of consumption reports
- Review of audits of compliance with antimicrobial prescribing guidelines

The ASP committee also works closely with the local microbiology service to extract and evaluate data on local antimicrobial resistance rates and changes or trends that emerge over time. This includes:

- Review of multi-drug resistant organism (MDRO) surveillance data every 3 months in liaison with a representative from the Infection Prevention and Control Committee;
- Review of cumulative hospital-level antibiograms every 12 months.

Targeted Interventions

Special initiatives or projects may be developed by the ASP Committee to target particular areas of practice. Examples of targeted interventions may include an 'IV to oral switch' project, an APS awareness campaign aimed at nursing staff, or development of a series of antimicrobial prescribing tutorials for medical officers.

References

Title of book/ journal/ articles/ Website	Author	Year of publication	Page
Antimicrobial Stewardship Programs in healthcare systems. Clinical Microbiology	MacDougall C, Polk R.	2005	638-656
Antimicrobial stewardship in Australian hospitals. Australian Commission on Safety and Quality in Healthcare: Sydney	Duguid M, Cruickshank M (eds).	2010	
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